

CHATEAURROUX AP, France

WMO: 073540

Lat: 46.870N

Lon: 1.741E

Elev: 515

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 34048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.0	(c) 25.6	(d) 16.9	(e) 13.2	(f) 25.1	(g) 20.8	(h) 15.9	(i) 28.6	(j) 31.2	(k) 51.5	(l) 27.8	(m) 49.4	(n) 7.6	(o) 80	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	90.8	68.5	86.6	67.4	82.8	66.0	70.7	85.2	69.1	82.9	67.6	79.7	7.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.0	75.3	64.7	93.3	73.4	63.3	88.7	72.0	34.8	85.4	33.5	83.0	32.3	80.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.5	(b) 21.0	(c) 18.5	(e) 17.9	(f) 97.0	(g) 5.2	(h) 4.0	(i) 14.1	(j) 99.9	(k) 11.1	(l) 102.2	(m) 8.2	(n) 104.4	(o) 4.4	(p) 107.3		
(a) 17.4	(b) 73.9	(c) 5.2	(e) 1.5	(f) 13.6	(g) 75.0	(h) 10.6	(i) 75.9	(j) 7.7	(k) 76.8	(l) 3.9	(m) 77.9	(n) 4.4	(o) 107.3	(p) 77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	40.1	41.3	46.7	51.2	57.8	64.7	68.4	68.0	61.5	55.2	46.1	41.0		
	DBStd	11.97	7.38	7.23	6.15	5.84	5.86	6.28	5.71	5.93	5.43	6.53	6.55	7.46		
	HDD50	1224	311	249	139	53	7	0	0	0	1	28	149	287		
	HDD65	4541	771	663	568	414	234	82	29	32	131	307	567	743		
	CDD50	2527	5	6	35	90	248	440	569	557	347	189	32	9		
	CDD65	368	0	0	0	0	10	71	133	124	27	3	0	0		
	CDH74	4085	0	0	0	27	154	801	1431	1315	320	37	0	0		
	CDH80	1459	0	0	0	1	19	271	540	544	81	3	0	0		
(14) Wind	WSAvg	8.5	9.7	9.8	9.5	8.6	7.9	7.7	8.1	7.6	7.5	7.8	8.7	9.6		
(15) Precipitation	PrecAvg	30.7	2.6	2.2	2.0	2.8	2.9	2.2	2.4	2.4	2.4	3.0	3.0	2.8		
	PrecMax	39.1	5.8	3.9	5.4	7.6	6.8	4.3	5.3	6.0	6.7	5.0	5.7	6.4		
	PrecMin	22.0	0.5	0.3	0.2	0.5	0.9	0.5	0.2	0.4	0.7	1.5	1.1	0.6		
	PrecStd	4.1	1.3	1.1	1.3	1.7	1.3	1.1	1.2	1.5	1.6	1.1	1.2	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	64.1	70.6	77.6	82.9	93.6	95.1	98.6	87.0	78.6	65.8	59.3		
		MCWB	51.9	51.1	55.8	61.9	66.3	70.2	69.2	69.5	65.4	63.5	57.6	53.0		
	2%	DB	54.8	57.6	65.8	72.6	78.6	86.9	90.0	91.4	81.8	73.3	61.1	55.9		
		MCWB	51.3	49.8	53.3	58.5	63.6	68.2	68.9	68.1	64.3	62.4	56.4	52.3		
	5%	DB	52.6	53.9	61.3	67.8	74.8	82.5	86.0	86.2	77.3	69.0	58.3	53.7		
		MCWB	49.6	49.1	51.7	56.2	62.3	66.7	67.3	67.3	63.1	60.6	54.6	51.0		
	10%	DB	50.3	51.5	57.5	63.3	70.6	78.3	82.1	81.3	73.3	65.4	55.8	51.7		
		MCWB	47.8	47.8	50.4	53.8	60.3	64.9	65.8	65.5	61.3	59.2	52.7	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	53.6	57.4	63.5	69.2	73.4	72.4	72.4	69.2	65.5	59.9	55.1		
		MCDB	55.8	60.2	67.6	74.9	79.0	87.1	87.8	87.7	79.3	73.7	63.3	57.2		
	2%	WB	52.0	51.2	54.8	59.8	65.9	70.1	70.5	70.2	66.6	63.5	57.1	53.0		
		MCDB	54.0	54.8	62.1	70.0	75.6	83.1	85.4	86.7	77.1	70.2	60.5	55.4		
	5%	WB	50.0	49.8	53.1	57.0	63.2	68.1	68.9	68.5	64.7	61.6	55.2	51.3		
		MCDB	52.1	53.3	59.4	65.8	72.3	79.7	82.7	82.7	74.2	67.5	57.9	53.4		
	10%	WB	48.0	48.2	51.3	54.7	60.9	66.2	67.2	66.8	62.8	59.8	53.0	49.5		
		MCDB	50.0	51.3	56.7	61.8	69.1	75.8	79.5	78.5	71.0	64.8	55.5	51.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	13.6	17.1	19.2	19.1	20.2	21.1	20.9	19.9	16.3	12.2	10.8		
		MCDBR	12.3	18.3	24.8	27.7	26.3	27.9	29.1	30.1	27.4	22.2	16.3	13.0		
		MCWBR	9.2	12.0	14.4	15.3	13.8	12.2	11.1	11.3	12.5	12.2	11.4	10.0		
	5% WB	MCDBR	10.8	15.3	20.4	24.8	23.6	25.2	25.4	25.0	21.8	19.1	14.8	12.2		
		MCWBR	9.0	10.8	12.6	14.3	13.3	12.4	10.5	10.6	11.0	11.1	11.1	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.336	0.364	0.395	0.397	0.410	0.402	0.394	0.362	0.358	0.330	0.313		
	taud		2.456	2.399	2.325	2.238	2.281	2.283	2.301	2.340	2.408	2.429	2.456	2.448		
	Ebn at Noon		246	263	271	272	275	272	270	269	251	237	232	232		
	Edh at Noon		22	28	34	41	41	41	40	37	31	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		346	613	999	1407	1650	1881	1832	1613	1255	762	418	298		
	RadStd		39	84	121	173	150	181	122	143	84	61	46	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page